Opening outcomes for training planning

Business Intelligence | C1 tracking report

Javier Alcaíno · Lucas Riquelme · Renato Varela
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Decision. Help a proposed club training coordinator choose opening families to review by player level, speed and color, using observed exposure and points with explicit denominators.

The case contains **86,950 unique eligible games**: 64,824 blitz and 22,126 rapid games from three short 2026 source windows. The most common checkpoint family is **Sicilian Defense** (9,255 games). These frequencies identify review material; they do not establish the best opening or prove that studying it improves performance.

Evidence	Value	What it means
Primary unit	86,950 games	One unique game per primary row
Opening checkpoint	20 plies	After Black's 10th move, using the observed prefix only
Rating coverage	400 to 3,066	Several focal-rating bands; blitz and rapid kept separate
Observed opening families	140	A fixed external position taxonomy, not a learned outcome label

Main finding and implication

White's overall descriptive points rate is 51.31% and draws account for 4.42% of the cohort. The relationship between rating difference and points is a material alternative explanation for an opening's score. For decisions, use color-specific, level-specific tables and their support counts; never pool both sides into an opening ranking.

Scope of this version

This is a reconstructed C1 baseline for tracking the team's new chess case. It does not replace or retroactively relabel the earlier school-audit submission. No instructor feedback or approval is invented. The report and notebook were requested without slides. Individual contributions and reflections await member confirmation.

The source covers only start-of-month UTC windows. It cannot represent full months, all Lichess users, or the effectiveness of an actual training intervention. No ML model or dashboard is implemented in C1.

Problem and decision map

The earlier project linked a concern about school expenditure to a documentary-review KPI, while several sources remained contextual rather than direct observations of the KPI's events. The new case seeks a closer alignment between decision, observed game, measurement and a feasible future predictive outcome. Public-data feasibility was compared for chess and Valorant; the team selected chess, openings, multiple levels, and separate blitz/rapid analyses.

The proposed user's need is a project assumption, not a claim from interviews. Official opening classifications and Lichess's public opening-explorer ecosystem establish an observable analytical context, not evidence that any particular club needs this intervention. No connection to a health policy or MINSAL numerical target is invented.

Seven-step map	Applied definition
1 Decision	A proposed coordinator prioritises opening-review material across speed, focal-player level and color.
2 Success criteria	Traceable, sufficiently supported review priorities. Actual learning gains need intervention and follow-up observations not available here.
3 Business question	How do opening exposure and observed points differ by rating band, speed and color, and can a later result be predicted from a fixed opening checkpoint?
4 Business process	Choose material, review games, hold a session, monitor later performance. This case observes games and results, not session delivery or learning.
5 Entities and events	Players participate in rated standard games, reach an observed 20-ply position, and later obtain a win, draw or loss.
6 Representation	One row per game; two descriptive color perspectives inherit the same game identity. Board positions map to a frozen opening lookup.
7 Measurement	Observed points per eligible game in the same color, speed, focal-rating band and opening family; include draws and disclose exclusions.

Alternatives considered

Timeout classification would have been feasible, but the team preferred openings for training. Rapid-only analysis would be simpler; the selected case instead retains both speeds while keeping their ratings and results separate. The older 2013-2017 convenience CSV was rejected in favor of recent official records, with an explicitly bounded extraction.

Measurement and opening definition

KPI = (wins + 0.5 × draws) / eligible games within a focal color × speed × player-rating band × checkpoint-opening family cell. Multiply by 100 to display observed points as a percentage. This differs from win percentage, which assigns no half-point to draws.

Component	Operational definition
Numerator	1 point for each win and 0.5 for each draw by the focal color.
Denominator	All unique completed games in that cell, reaching a valid nonterminal checkpoint. Draws stay in the denominator.
Period	Only the observed start-of-month windows in July, August and September 2026, not complete months.
Levels	Under 1200; 1200-1599; 1600-1999; 2000+. These are analysis conventions applied to the focal player's Lichess rating.
Display rule	At least 50 games for highlighted heatmap cells. Small cells are retained in the complete tables; 50 is not a precision guarantee.
Interpretation limit	A review signal based on historical observations, not a causal effect or measured training benefit.

Why both colors must remain separate

One game's two perspective scores sum to 1. If both sides are pooled, each opening family's combined mean is mechanically 50%. The primary database keeps one row per game; a long table supports separate color comparisons without pretending that two perspectives are independent games.

A family available at the prediction checkpoint

The fixed reference contains 3,865 named positions. Starting from the initial board, the analysis replays only the first 20 observed plies and retains the last named position actually encountered. The family is the part of its English name before a colon. Position matching handles transpositions. There are 0 equal-position naming ties in this reference; the documented fallback rule is outcome-independent lexicographic order.

The process is many observed game prefixes to one fixed lookup entry at a matching board position. Unknown families would be retained as Unclassified. The player's result, final opening header and later moves are not used to name the checkpoint opening.

In 311 games, the source's final family differs from the prefix family. Later moves and taxonomy differences can both contribute. The columns are therefore not interchangeable. Games not reaching the checkpoint or with an already terminal board there are excluded transparently.

Sources and coverage

The input comes from official Lichess standard-rated PGN archives under CC0. For each month, exactly the first 16 MiB of compressed bytes were downloaded with verified HTTP 206 range responses. The final uncertified PGN record in each prefix was discarded. This provides a small reconstructible input, rather than requiring three complete monthly archives.

Archive month	Source records	Earliest recorded start UTC	Latest recorded start UTC
2026-07	51,478	07.01 00:00:00	07.01 00:43:20
2026-08	51,419	08.01 00:00:00	08.01 00:36:21
2026-09	50,920	09.01 00:00:00	09.01 00:40:14

The timestamps describe source-recorded game starts, not exact game-end times or proof of complete coverage inside each interval. The sampling mechanism is a deterministic archive-prefix availability window. Compression ratios and game content produce unequal window lengths.

Population layer	Meaning
Intended	Rated standard blitz/rapid play relevant to training planning across levels.
Observed	All fully bounded source records in the three byte prefixes, including out-of-scope speeds before filtering.
Analytical	86,950 valid unique selected-speed games reaching the 20-ply nonterminal checkpoint.

Selection limits

The case excludes other times of day, later days and other platforms. Selection into rated Lichess play differs from participation in a particular club. Repeated users contribute multiple games. Surviving to the checkpoint excludes short games and changes the target population. A random sample drawn within this case cannot fix those coverage limits.

Source preservation and privacy

Original byte prefixes, response headers, reduced source CSV and SHA-256 checksums are retained. Public usernames are replaced by stable pseudonyms in the reduced CSV; game IDs remain public identifiers, so anonymity is not guaranteed. The opening reference is frozen to a recorded Git commit. Its October 2026 snapshot supports retrospective analysis, not a claim that exactly this taxonomy was deployed in July.

Preparation and row accounting

All transformations are deterministic scope and validity rules. No threshold was chosen to improve a model score. Original inputs remain unchanged. No imputation or outlier deletion was introduced merely to make the workflow look more elaborate.

Rule	Before	After	Excluded
One record per game ID	153,817	153,817	0
Select blitz and rapid	153,817	91,068	62,749
Valid UTC date/time	91,068	91,068	0
Completed allowed result	91,068	91,068	0
Numeric plausible ratings	91,068	91,068	0
Valid game/player IDs	91,068	91,068	0
Clock syntax and speed consistency	91,068	91,068	0
Reached 20 observed plies	91,068	86,997	4,071
Legal, nonterminal checkpoint	86,997	86,950	47

The rating range 1-4000 is a broad plausibility convention, not an asserted official rating boundary. Clock estimates use base seconds + 40 × increment, with Lichess blitz at 180-479 seconds and rapid at 480-1499 seconds. Valid extreme ratings are retained.

Checks applied to the prepared cohort

Check	Flagged cases
Raw duplicated game IDs	0
Conflicting payloads within duplicated IDs	0
Prepared game ID uniqueness	0
Required identifiers missing	0
Numeric ratings and plausible range	0
Allowed result and score mapping	0
Valid source timestamps	0
Observed month matches archive month	0
Legal 20-ply prefix and live board	0
Unknown opening family	0
Clock/category consistency	0

A zero count is evidence of the stated check only, not proof of exhaustive data correctness. The exact failure actions and raw profiling table remain visible in the notebook. Exclusions of other speeds and games before the checkpoint are scope changes, not claims that those records are erroneous.

Five Cs and descriptive safeguards

Lens	Evidence in this investigation
Clean	Check source identifiers, duplicates, numeric ratings, dates, allowed outcomes, clock format and legal prefixes. Keep valid extremes and explicitly label unknowns.
Consistent	Use the same point formula, checkpoint, family lookup and speed boundaries across all windows. Maintain speed-specific rating interpretation.
Conformed	Match named board positions to one frozen reference; record lookup coverage and tie rules. Assert exactly two complementary perspective scores per game.
Current	July-September 2026 games were available before the October analysis. Recent records do not remove limited temporal coverage.
Comprehensive	Retain the selected speeds and observed levels; disclose missing UTC periods, other platforms, short games, player histories and unobserved interventions.

Numerator and denominator safeguards

Exposure counts are unique games. Color-specific KPI cells count a game once for the relevant color. Draws contribute half a point rather than disappearing. Small heatmap cells are masked because they fail the display-support rule, not because their observed score is zero. Complete exported tables retain those cells.

Association safeguards

The following figures describe distributions, relationships and segment comparisons. They do not estimate a causal benefit from changing an opening. Opponent strength, selection into an opening, repeated players and later decisions can explain score differences. No independence-based confidence intervals or significance claims are used.

Coverage of the opening lookup

The analytical cohort contains 140 distinct family names and 0 Unclassified records. Full classification coverage does not mean every family is equally specific: some names remain broad and may match an early generic position. The last matched ply and full matched name are retained for audit.

Scope of data integration

The join is an outcome-independent position lookup rather than a merge of several fact tables with incompatible grains. A final source tag remains an audit column. It is not silently substituted when a prefix lookup fails, and no global opening win rate is encoded into a predictive input.

Ratings and observed population

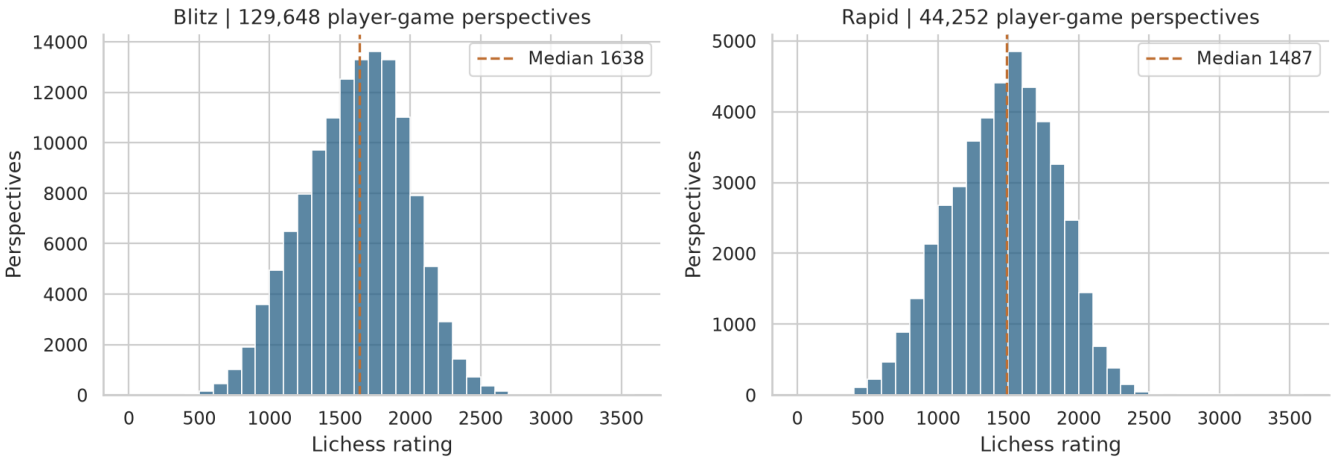


Figure 1. Rating distributions by speed. Dashed lines mark medians. Each plotted item is a player-game perspective, including repeated players.

Speed	Perspectives	Mean	Median	Q1	Q3
blitz	129,648	1615.9	1638	1363	1879
rapid	44,252	1465.4	1487	1207	1732

The distribution supports segmenting review material by observed rating rather than assuming one homogeneous audience. Mean and median describe different aspects of the distribution; neither is a measure of improvement. Speed-specific ratings should not be treated as a common cross-speed ability scale.

Repeated players contribute more than one perspective. The histogram therefore describes the distribution of observed play opportunities, not the distribution of distinct club members or all Lichess users. The apparent range cannot repair the window-selection mechanism.

Opening exposure and review material

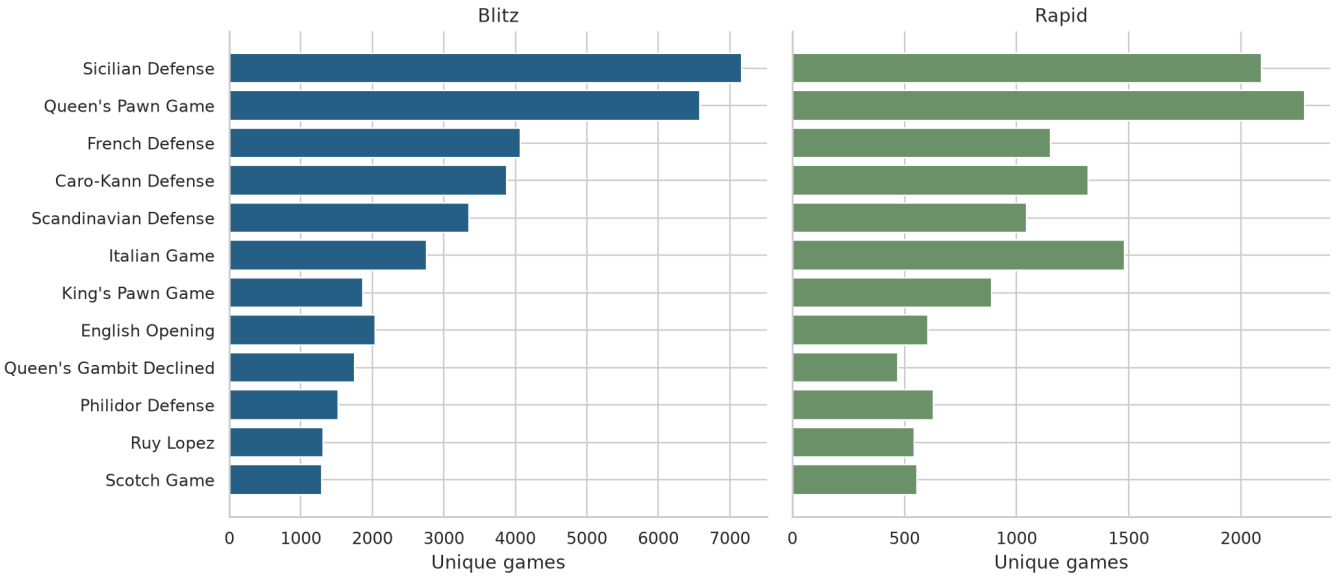


Figure 2. The 12 most frequent checkpoint families, with separate speed counts. Family selection is by exposure, not outcome.

Speed	Family	Games	Share within speed
blitz	Sicilian Defense	7,164	11.05%
blitz	Queen's Pawn Game	6,584	10.16%
blitz	French Defense	4,071	6.28%
blitz	Caro-Kann Defense	3,875	5.98%
rapid	Queen's Pawn Game	2,284	10.32%
rapid	Sicilian Defense	2,091	9.45%
rapid	Italian Game	1,482	6.70%
rapid	Caro-Kann Defense	1,320	5.97%

High exposure makes a family a plausible source of relevant examples. Frequency does not prove that the opening deserves more training than a rarer weakness, nor that its observed score is caused by the opening. Review priority still depends on level, color, support and the coach's inspection of the games.

Rating differences and points

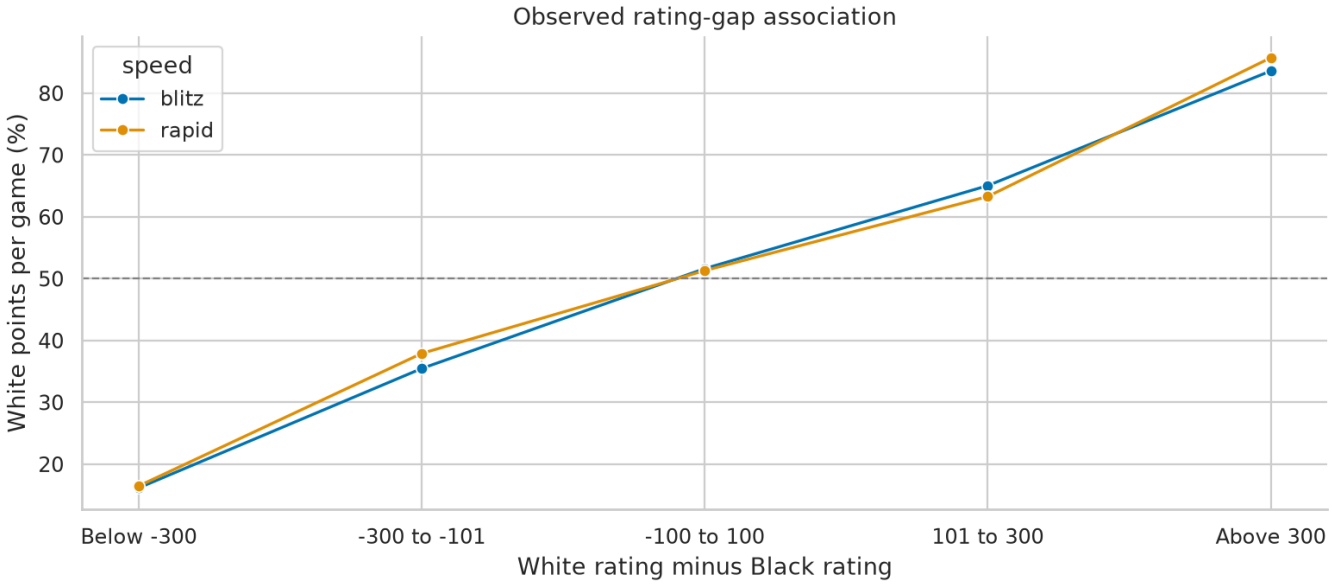


Figure 3. White points per game by rating-gap category, separately by speed. Counts and discrete category boundaries are shown below.

Speed	White minus Black rating	Games	White points
blitz	Below -300	1,184	16.13%
blitz	-300 to -101	3,555	35.44%
blitz	-100 to 100	55,355	51.60%
blitz	101 to 300	3,583	65.02%
blitz	Above 300	1,147	83.57%
rapid	Below -300	462	16.45%
rapid	-300 to -101	1,522	37.88%
rapid	-100 to 100	18,190	51.28%
rapid	101 to 300	1,501	63.26%
rapid	Above 300	451	85.70%

The game-level Spearman association between rating gap and White points is 0.149. The relationship is descriptive, but it motivates treating opponent strength as an alternative explanation for opening scores. It is not an estimate of a causal rating effect or a calibrated result probability.

W2 continues this observation with a declared close-rating restriction, retaining the same KPI and explicitly comparing the changed population. C1 does not select a predictive model or choose thresholds based on future performance.

Blitz points by opening and level

Every visible cell shows points (%) and its number of unique games. Blank cells have fewer than 50 games. Scores remain unadjusted observational summaries; compare within focal level, color and speed.

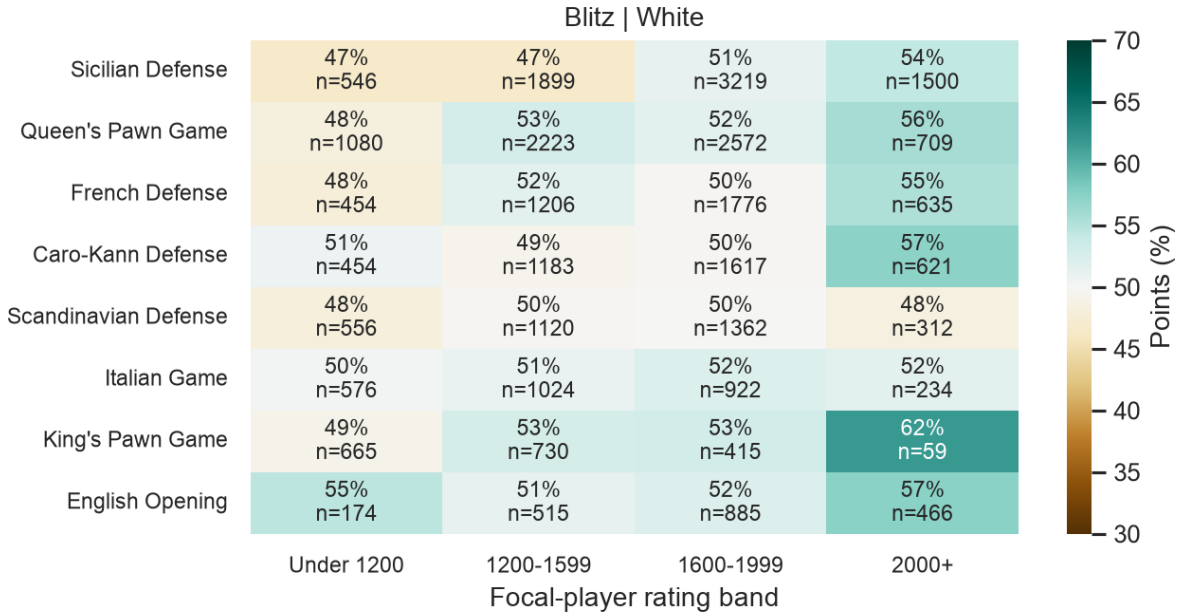


Figure 4bw. Blitz, white perspective. The same frequent families are displayed to support comparison rather than selecting high-scoring winners.

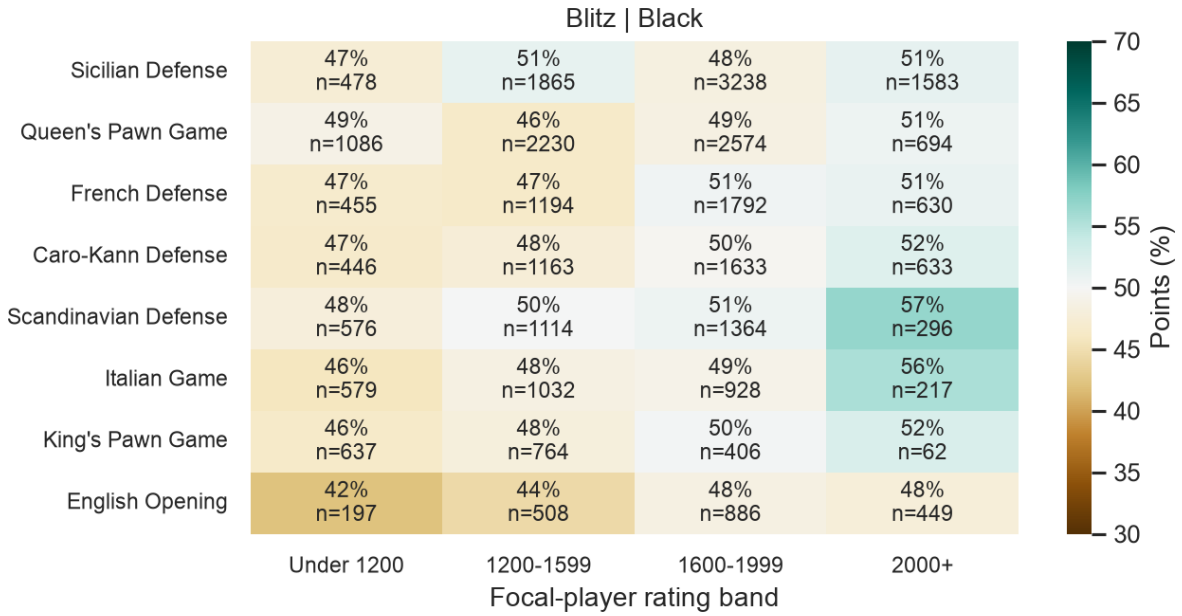


Figure 4bb. Blitz, black perspective. The same frequent families are displayed to support comparison rather than selecting high-scoring winners.

The complementary colors must not be averaged into a universal family score. A cell with a high score may contain stronger focal players within the band or a selected subset of opponents. Large support improves descriptive stability but does not make the comparison causal.

Rapid points by opening and level

Every visible cell shows points (%) and its number of unique games. Blank cells have fewer than 50 games. Scores remain unadjusted observational summaries; compare within focal level, color and speed.

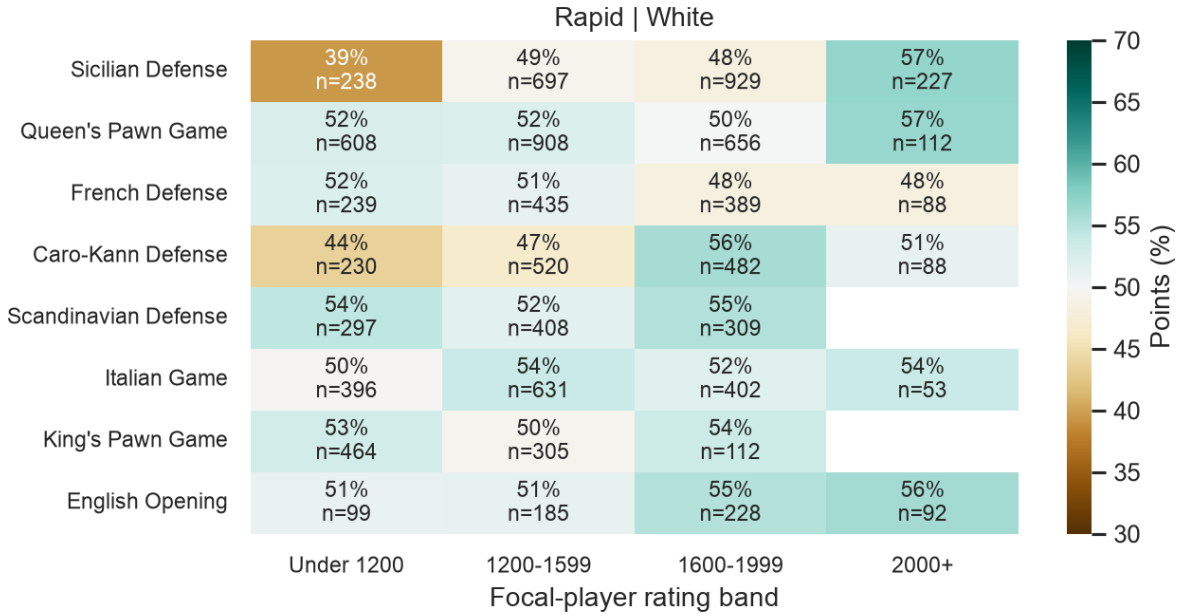


Figure 4rw. Rapid, white perspective. The same frequent families are displayed to support comparison rather than selecting high-scoring winners.

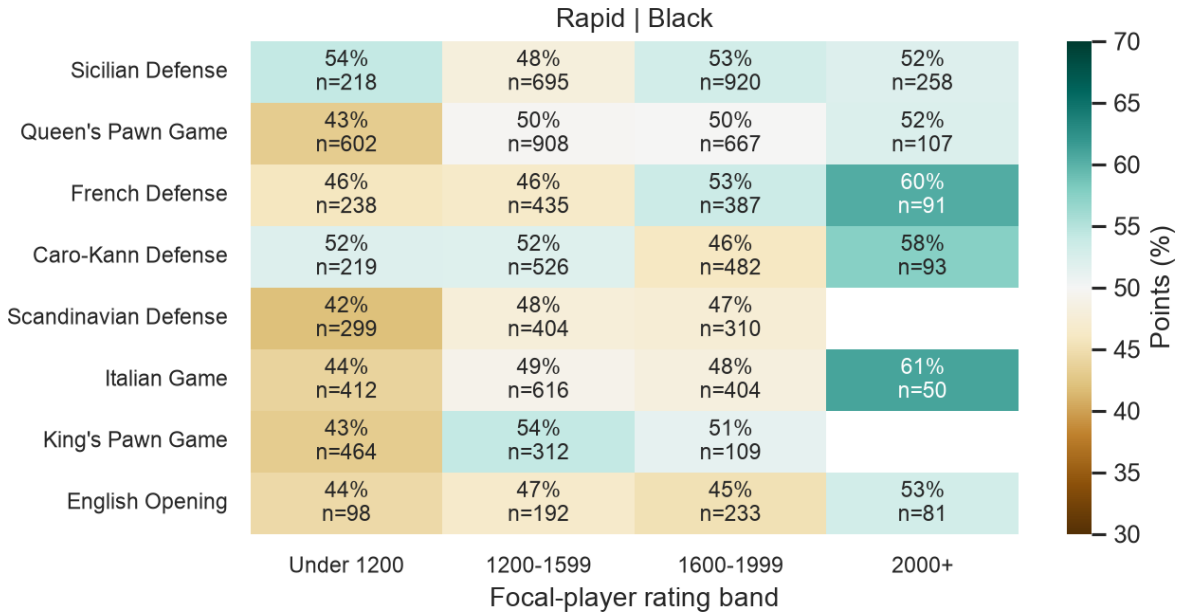


Figure 4rb. Rapid, black perspective. The same frequent families are displayed to support comparison rather than selecting high-scoring winners.

The complementary colors must not be averaged into a universal family score. A cell with a high score may contain stronger focal players within the band or a selected subset of opponents. Large support improves descriptive stability but does not make the comparison causal.

Provisional training agenda

This agenda chooses the most frequent supported family in each focal level, speed and color. It is an exposure-led rule for selecting relevant examples, not a ranking of opening quality. All choices require at least 50 observed games; alphabetical family order resolves exact count ties without consulting results.

Speed	Color	Rating band	Review family	Games	Points
blitz	black	Under 1200	Queen's Pawn Game	1,086	48.8%
blitz	black	1200-1599	Queen's Pawn Game	2,230	46.5%
blitz	black	1600-1999	Sicilian Defense	3,238	48.4%
blitz	black	2000+	Sicilian Defense	1,583	51.3%
blitz	white	Under 1200	Queen's Pawn Game	1,080	48.2%
blitz	white	1200-1599	Queen's Pawn Game	2,223	52.7%
blitz	white	1600-1999	Sicilian Defense	3,219	51.2%
blitz	white	2000+	Sicilian Defense	1,500	54.3%
rapid	black	Under 1200	Queen's Pawn Game	602	43.0%
rapid	black	1200-1599	Queen's Pawn Game	908	49.7%
rapid	black	1600-1999	Sicilian Defense	920	53.1%
rapid	black	2000+	Sicilian Defense	258	52.1%
rapid	white	Under 1200	Queen's Pawn Game	608	52.4%
rapid	white	1200-1599	Queen's Pawn Game	908	52.3%
rapid	white	1600-1999	Sicilian Defense	929	48.3%
rapid	white	2000+	Sicilian Defense	227	56.8%

The points column provides outcome context for the observed examples. A lower score can motivate closer inspection, but neither the agenda nor the score proves an opening caused a result. The same family can appear for multiple audiences; the representative games and teaching depth would differ by level and color.

The coordinator's proposed next action is to inspect representative games in these cells before choosing lesson content. Actual opening knowledge, session costs and training effects are unobserved. W2 preserves this agenda and reports how it changes when the same selection rule is applied to close-rating opponents.

Missingness and diagnostic association

Source variable	Empty strings	Null values
white_rating_diff_post	481	0
black_rating_diff_post	481	0
prefix_san_20	394	0

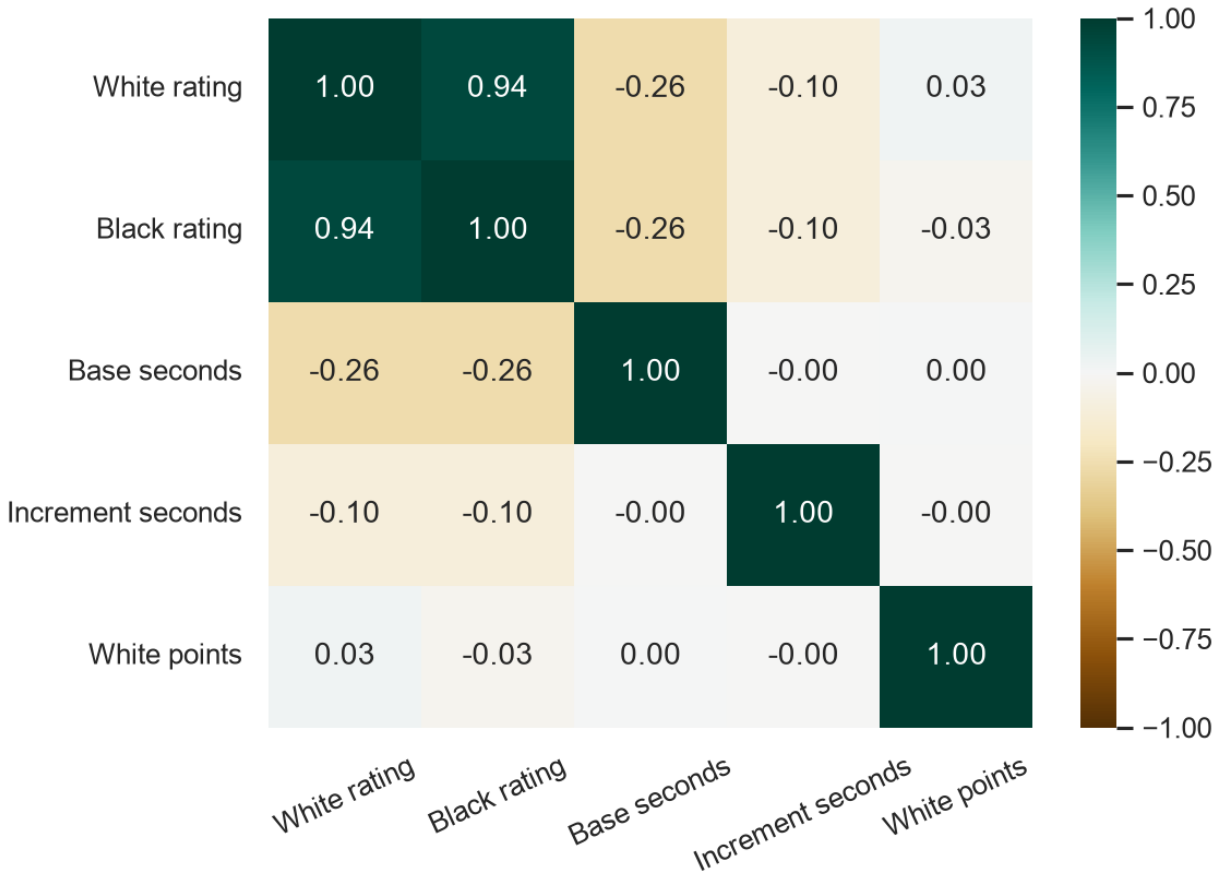


Figure 5. Spearman associations in the unique-game cohort. Ratings, configured clock parameters and White points are shown for diagnostic interpretation only.

Positive or negative correlations describe association in these windows. A small coefficient does not prove absence of an effect. The relationship between the two ratings also reflects who gets paired, not an intervention. The analysis does not turn descriptive correlations into a causal explanation or a promise of predictive performance.

Full-cohort exploration is legitimate for this C1 baseline. It also means a later holdout from this same cohort has already been observed. W2 must disclose that limitation instead of calling its test subset perfectly untouched.

Feasible continuation and limitations

A concrete supervised task

A future model could predict three-class game result - White win, draw or Black win - immediately after Black's 10th move. Available inputs are the two ratings, their difference, configured clock parameters, speed and the prefix-derived family. The target is a later observed game outcome. This is a feasible information contract, not a performance claim.

Use	Available inputs or required boundary
Training planning	Historical family exposure, color-specific points and denominators by level and speed.
Future prediction	Only checkpoint-available fields; no final tags, post-game rating changes, complete move counts, termination or result.
Future dashboard	Filters for speed, level, color, family and observed source window; display counts, points, coverage and quality flags.
Evidence still missing	Training interventions, learning gains, wider calendar coverage and genuinely new evaluation data.

What the analysis cannot establish

The source does not observe why a player chose an opening, actual opening knowledge, coaching exposure or learning outcomes. Later middlegame/endgame decisions also affect the final result. A broad family can contain very different variations. The source windows are not representative months, and the checkpoint cohort omits shorter games.

Repeated players and opponents create dependence. The intended future-play task is not an unseen-player task; separating only White's ID would not prevent the same person appearing as Black. A separate both-opponents grouping design would be needed for a new-player generalisation claim.

Bounded recommendation

Use common, adequately supported cells to select examples for human review, then compare within speed, color and focal level. A low observed score is a reason to inspect games, not a verdict that an opening is intrinsically bad. Do not assert a universal best opening or numerical improvement from training. Validate data and expand coverage before evaluating a simple model.

Process record

Decision	Alternative and verification
Team scope	The team chose openings rather than timeout prediction; selected multiple levels and both speeds. Those choices are reflected in the measurement contract.
Recent reduced data	Rejected the older convenience CSV. Verified official 2026 URLs, HTTP byte ranges, source hashes and exact retained record counts.
Prefix measurement	Rejected final source opening tags as predictors. Replayed legal observed prefixes, checked lookup cardinality and retained mismatches for audit.
KPI definition	Included draws as half a point; retained denominators and color separation. Asserted one game and two complementary descriptive views.
Quality decisions	Kept valid extremes; excluded out-of-scope and invalid checkpoint rows transparently. Did not invent nulls, imputation or required joins.
Continuation	Defined a checkpoint-available target/input contract. No model fitting or performance promise was added to C1.

Instructor feedback and response

The user stated that C1 feedback was not yet available and would refer to the former topic. No instructor comments or approval are fabricated. W2 records explicit self-audits and preserves this baseline so that subsequent changes can be inspected.

Individual contributions and reflection

Confirmed members are Javier Alcaíno, Lucas Riquelme and Renato Varela. The group supplied its scope preferences through this conversation. Individual contributions, learning gains and unresolved uncertainties have not been provided and remain to be confirmed by each member in TEAM_REVIEW.md. This tracking report does not invent personal statements or claim a complete historic resubmission.

AI use and verification

AI assisted research, code and writing. The user selected openings over the suggested timeout topic, both speeds over rapid-only, and recent over historical data. Verification used official hashes/ranges, actual row accounting, legal prefix reconstruction, KPI denominator checks and fresh-kernel notebook execution. These checks do not establish that every member has already reviewed or can defend all outputs.

References and reproduction

1. **Lichess open database**. Official standard-rated July, August and September 2026 archives; CC0; byte prefixes retrieved 21 October 2026.
https://database.lichess.org/#standard_games

2. **Lichess opening names**. CC0 reference, frozen commit a6189a30dc273ccb21fc2536a9a2fefd5592a67a. Family convention and position matching.
https://github.com/lichess-org/chess-openings

3. **Lichess FAQ**. Configured-clock speed classification and platform context.
https://lichess.org/faq

4. **scikit-learn common pitfalls**. Training-only learned preprocessing and leakage prevention for the W2 continuation.
https://scikit-learn.org/stable/common_pitfalls.html

5. Fontecilla Correa, Tomás. Business Intelligence course materials supplied by the team: C1 IIB423T-1 brief and rubric; Workshop 2 student brief and rubric; Classes 01-04 and course calendar. Exact source filenames and SHA-256 values are in provenance/course_sources.json. W1 scope was reconstructed from C1, the calendar and prior work; no unavailable W1 rubric is reproduced.

Inspectable evidence

The report contains the substantive method and findings. notebooks/C1_openings.ipynb contains the executed preparation, profiling, quality checks, five figure groups and interpretation. data/raw/ contains the unchanged reduced source; data/processed/ contains the prepared cohort. outputs/tables/ contains all KPI cells, including small cells omitted from heatmaps.

provenance/source_manifest.json records byte ranges, record counts, dates, source hashes and the fixed opening taxonomy. scripts/acquire_data.py reconstructs the reduced input from supplied prefixes or public ranges. scripts/build_report.py rebuilds this PDF from the saved C1 results. requirements.txt and README.md document software and execution order.

Execution evidence: C1 ran in a fresh Python 3.12.14 kernel, top to bottom, with 14 executed code cells, 5 saved figure outputs and 0 recorded errors.

Deliverable boundary: local tracking package; no slide deck, repository publication, assessment submission or personal sign-off is claimed. Earlier school-audit work remains unchanged in its original location.